NEXT-GEN PROJECT
REPORT ON
"SAMARPAN"

Submitted in partial fulfillment of requirement for the award of the degree of

MASTER' OF COMPUTER

APPLICATIONS SCHOOL OF

COMPUTER APPLICATIONS

SESSION: 2025-2027

SUBMITTED BY:

SAHIL PRATEEK TIGGA (12521526)
SAKSHAM SHAKYA (12508547)

SUPERVISED BY:

YOGESH

ASSISTANT PROFESSOR SCHOOL OF COMPUTER APPLICATIONS

LOVELY PROFESSIONAL UNIVERSITY,

JALANDHAR, PUNJAB - 1444

TABLE OF CONTENTS

1. Introduction

Samarpan is a comprehensive blood donation management platform that connects donors with patients needing blood or platelets in real-time.

Built with modern web technologies such as Next.js, TypeScript, MongoDB, Tailwind CSS, and Redux Toolkit, the system provides a secure, scalable, and mobile-first solution.

The platform supports secure authentication, donor tracking, event management, and an adaptive admin dashboard designed for real-world emergency usage.

2. Project Overview

Samarpan streamlines the complete blood donation lifecycle—from donor registration to successful transfusion support.

- Real-time donor–patient matching
- Donation history tracking and digital records
- Certificate generation for participants
- Priority-based emergency blood requests
- Reliable QR code scanning for events
- Bulk notifications and communication tools
- Full maintenance system with route protection and IP whitelisting
- Mobile-first responsive design with adaptive UI

3. Objectives

Samarpan aims to create a real-time, mobile-first platform that efficiently connects blood donors with patients in urgent need of blood or platelets, saving lives by eliminating traditional barriers in the donation process.
The core objective is to streamline the entire blood donation ecosystem—from donor registration and matching to event management and certification—through a reliable, scalable web application accessible anytime, anywhere, particularly during emergencies when users are likely on mobile devices.
This addresses critical gaps in current systems by providing intuitive tools for donors (history tracking, instant alerts), patients (priority requests), and administrators (QR check-ins, maintenance controls), ultimately fostering faster human connections that turn willingness into lifesaving action.

- Connect blood donors and patients in real-time
- Provide a mobile-first platform for emergency accessibility
- Streamline donor registration, tracking, and verification
- Enable priority-based urgent blood requests
- Simplify event management and participation
- Provide administrators with powerful operational tools
- Promote voluntary donations and equitable blood access

4. Literature Review

India faces a persistent blood shortage with demand significantly exceeding supply, creating critical healthcare challenges.

- Large annual deficit in blood unit collection
- Seasonal shortages worsening supply-demand imbalance
- Fragmented systems across blood banks and hospitals
- Manual processes causing delays and matching inefficiencies
- Low voluntary and repeat donor rates
- Limited centralized platforms for real-time coordination

Existing systems lack mobile-first urgency features, real-time synchronization, and comprehensive administrative tools.

Samarpan addresses these gaps through a human-centered, technology-driven platform focused on speed, reliability, and accessibility.

5. System Features

5.1 Donor & Patient Tools

- Secure signup and authentication
- Personal profile dashboard
- Real-time matching alerts
- Donation history tracking
- Blood request creation with priority levels
- Event participation and auto-generated certificates

5.2 Admin & NGO Capabilities

- Centralized management dashboard
- User, donation, and event management
- QR code scanning and check-in system
- Transportation coordination tools
- Bulk notification system
- Comprehensive maintenance and monitoring controls
- Adaptive mobile-first admin interface

6. System Setup & Deployment

- Clone repository and project setup
- Install dependencies using npm
- Configure environment variables
- Initialize database and super admin setup
- Run development server for local deployment

7. Conclusion

Samarpan transforms the blood donation ecosystem by turning willingness into lifesaving action through real-time technology.

Its mobile-first architecture, intelligent matching, and powerful administrative tools make it a reliable and scalable platform for modern healthcare support